

assignment 8 - magnetic field, magnetic force

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:12pm

Quiz Instructions

attempts 2. Keep the last one. I did a tutorial (see dropbox). But try first without hints.

READ before:

A magnetic field can deflect a moving charge but can not change its speed or KE.

A magnetic field can deflect a moving charge only if there is a component of the velocity perpendicular to B.

A magnetic field loops from N pole to S pole.

Check your equation sheet:

A moving charge with a velocity V inside a magnetic field is acted upon by a force = $qVB\sin(\alpha)$
 α is the angle between V and B .

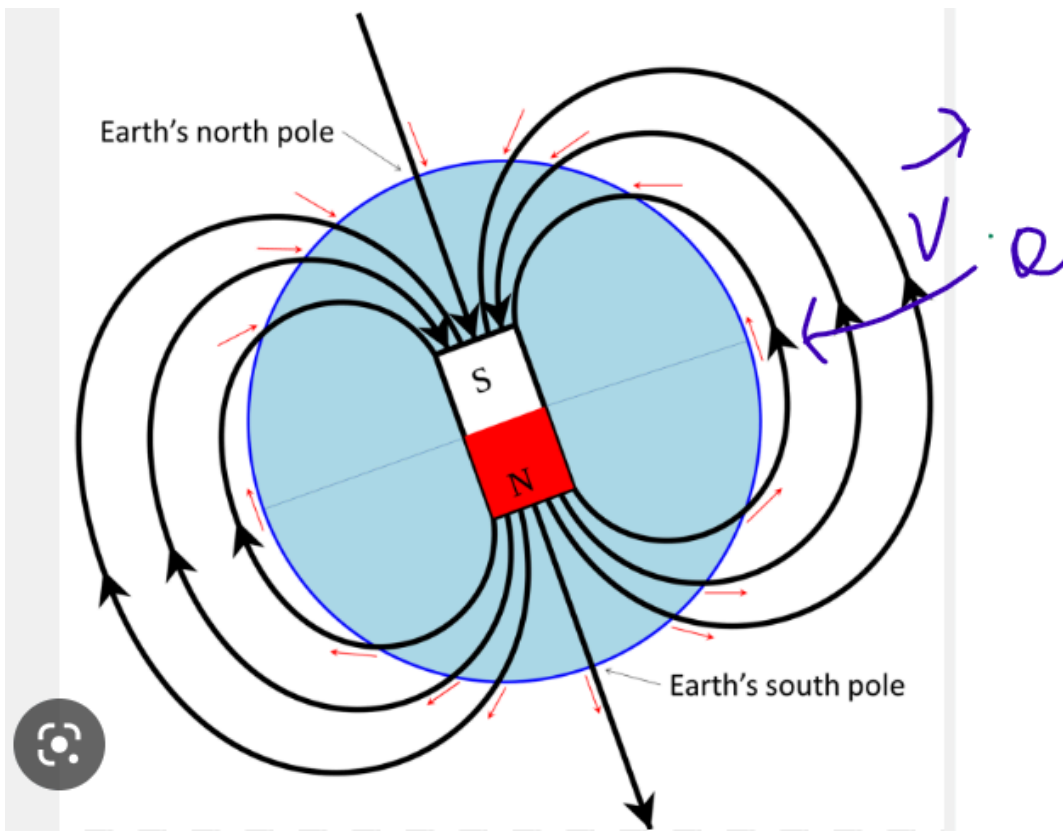
A wire carrying current I of length L inside a magnetic field is acted upon by a force = $IL\sin(\alpha)$. α is the angle between I and B

$\tau = N I A B \sin \theta$. This is the torque on a current-carrying loop in a uniform magnetic field. This equation can be shown to be valid for a loop of any shape. The loop carries a current I , has N turns, each of area A , and the perpendicular to the loop makes an angle θ with the field B



Question 1 3.5 pts

A geophysicist measures the magnetic force exerted on a proton that is moving vertically toward the Earth (blue vector V in the figure below. e is the proton) at a point 1.996 km/s at Earth's equator. At that location, Earth's field is horizontal and parallel to ground (see the small red arrows or field lines) and has a magnetic field B magnitude of 0.40×10^{-4} T. What are the magnitude and direction of the force she will measure? ($e = 1.60 \times 10^{-19}$ C). Use the right hand rule



hint: the magnetic force on a moving charge inside a magnetic field B is $F = qVB$ (in magnitude). V is the speed of the charge. this equation works when V and B are perpendicular. Use the right hand rule for the direction of the force (its V , B , F). See tutorial. Also see your equation sheet.

☐

1.3 10^{-20} N toward North

☐

1.3 10^{-20} N out of the screen

☐

1.3 10^{-20} N into the screen

⋮

Question 2 3.5 pts

A beam of electrons passing through a magnetic field experiences **maximum** deflection if the direction of the beam is

hint: $F = q V B \sin(\theta)$ with θ the angle between V (speed of the particles) and B (magnetic) field). A beam of electrons are electrons moving at a given speed V .

☐

parallel to the field lines.

☐

perpendicular to the field lines.



none of these



Question 3 3.5 pts

The direction of the force exerted on a moving charge in a magnetic field is



in the direction of the motion.



opposite its motion.



at right angles to the direction of the motion.



Question 4 3.5 pts

A proton moving at 5.0×10^4 m/s horizontally enters a region where a magnetic field B field of 0.12 T is present, directed vertically downward. What magnitude force acts on the proton due to this field? ($e = 1.60 \times 10^{-19}$ C)

hint: just apply $F = qVB$.



4.2×10^{16} N



9.6×10^{-16} N



3.5×10^{-12} N



Question 5 3.5 pts

Whereas electric charges can be isolated (monopoles), magnetic poles



gather in clusters.



can also.



cannot.



none of the above



Question 6 3.5 pts

A beam of electrons can pass through a magnetic field without being deflected if the direction of the beam is

☐

none of these

☐

parallel to the field lines.

☐

perpendicular to the field lines.



Question 7 3.5 pts

A straight wire that is 0.60 m and carrying a current of 2.0 A is placed at an angle with respect to the 0.30-T magnetic field B field. If the magnetic force exerted on the wire has a magnitude of 0.18 N, what angle does the wire make with respect to the magnetic B field?

hint: Check equation sheet. $F = IlB\sin(\theta)$ θ is the angle between I and B

☐

60°

☐

35°

☐

30°

☐

20°

☐

25°



Question 8 3.5 pts

Like kinds of magnetic poles repel while unlike kinds of magnetic poles

☐

repel also.

☐

attract.

☐

may attract or repel.



Question 9 3.5 pts

The intensity of cosmic rays bombarding the Earth's surface is most at the

☐

poles.

☐

equator.

☐

mid-latitudes.



Question 10 3.5 pts

A proton travels at a speed of 5.0×10^7 m/s through a 1.0-T magnetic field. What is the magnitude of the magnetic force exerted by the field on the proton if the angle between the proton's velocity and the magnetic field B vector is 30° ? ($e = 1.60 \times 10^{-19}$ C)

hint: equation sheet. see tutorial. $F = q v B \sin(\theta)$



4.0×10^{-14} N



4.0×10^{-12} N



2.0×10^{-12} N



2.0×10^{-14} N



Question 11 3.5 pts

As the number of loops in a current-carrying wire is increased, the



more spread out is the magnetic field.



weaker the current.



stronger the enclosed magnetic field.

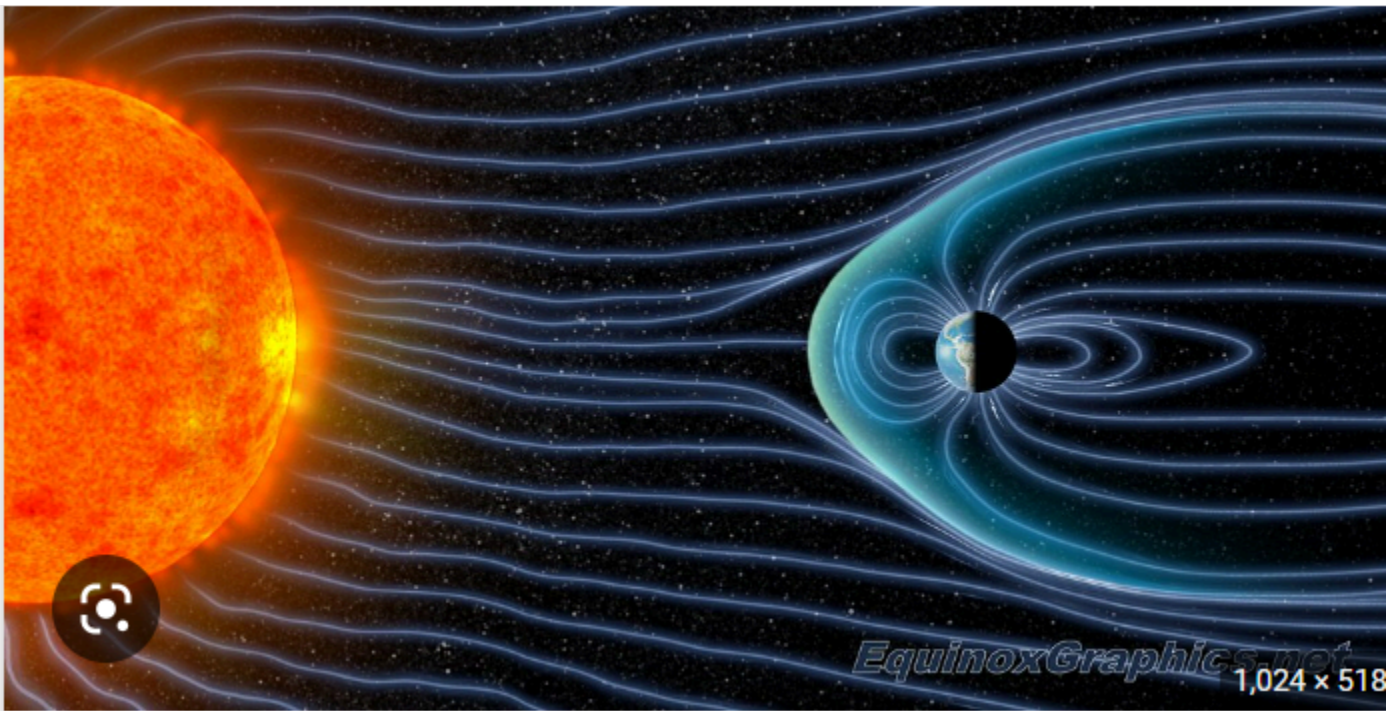


greater the back emf.



Question 12 3.5 pts

Solar winds headed in Earth's direction are deviated mainly by



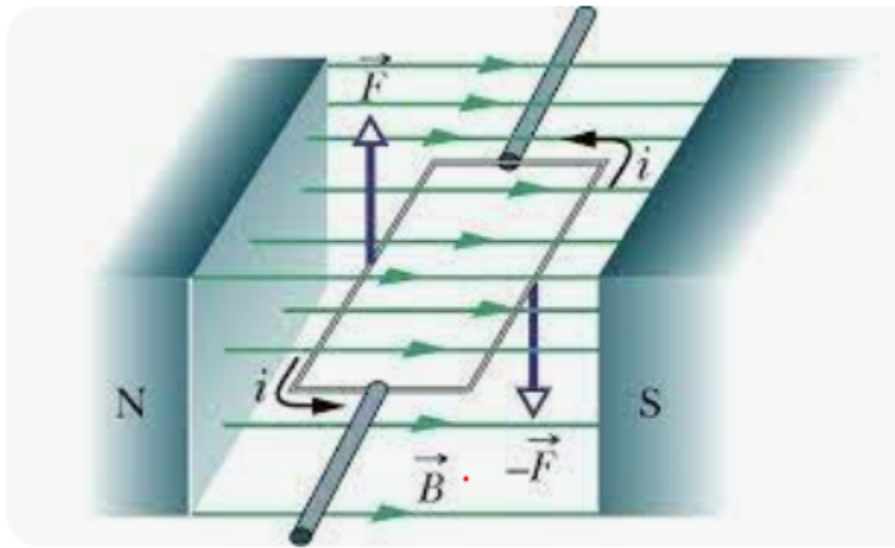
- ☐ Earth's magnetic field.
- ☐ Earth's radiation belts.
- ☐ the troposphere.
- ☐ the upper atmosphere.



Question 13 3.5 pts

A flat circular coil of wire having 200 turns and diameter 6.0 cm carries a current of 7.0 A. It is placed in a magnetic field B field of 0.60 T. What is the magnitude of the magnetic torque exerted on the coil? hint: see slide.

torque = number loops x area loop x $I \times B$



Steemit

- ☐ 1.2 N · m
- ☐ 3.9 N · m
- ☐ 2.4 N · m
- ☐ 1.0 N · m
- ☐ 5.4 N · m



Question 14 3.5 pts

A current-carrying coil of wire is

- ☐ an electromagnet.
- ☐ a superconducting magnet.
- ☐ a superconductor.
- ☐ a magplane in the making.



Question 15 3.5 pts

The conventional direction of magnetic field lines outside a magnet are from

- ☐ north to south.
- ☐ south to north.



either way



Question 16 3.5 pts

If you break a bar magnet in half you'll



have two magnets.



have two and a half magnets.



destroy its magnetic properties.



none of the above



Question 17 3.5 pts

A thin copper rod 1.0 m long has a mass of 0.050 kg and is in a magnetic field B field of 0.10 T. What minimum current in the rod is needed in order for the magnetic force to balance the force exerted on the rod by Earth?

hint: the weight is mg and is balanced by the magnetic force ILB



2.5 A



1.2 A



4.9 A



9.8 A



Question 18 3.5 pts

Pigeons navigate primarily by



magnetic sensors in their heads.



a keen sense of smell.



ultra-high-pitched sounds.



a good memory.



none of the above



Question 19 3.5 pts

Although a magnet can change the direction of travel of an electron beam, it cannot change its



I can not change its speed but can change its KE



It can change the speed or the KE



It can not change its KE nor its speed



it can not change its KE but can change its speed



Question 20 3.5 pts

A moving charge inside a magnetic field will experience a magnetic force (qVB) that will deflect the charge. If the magnetic field is perpendicular to the velocity of the charge, the charge will describe circle. It will circle the magnetic field at a constant speed. The magnetic force is a centripetal force.

A beam of singly-charged ions entering a magnetic field are swept into circular paths. The wider paths are those of

hint: (m is the mass of the particle. V is the speed. q the charge. r the radius of the circle described)



the lighter ions.



neither particularly.



the heavier ions.



Question 21 3.5 pts

Surrounding every magnet is



an electric field.



a magnetic field.



neither of these



both of these



Question 22 3.5 pts

Surrounding every stationary electron (so at rest) is

- ☐ an electric field.
- ☐ both of these
- ☐ neither of these
- ☐ a magnetic field.



Question 23 3.5 pts

Which force field can increase a moving electron's speed?

- ☐ neither of these
- ☐ electric field
- ☐ magnetic field
- ☐ both of these



Question 24 3.5 pts

The source of all magnetism is

- ☐ ferromagnetic materials.
- ☐ tiny domains of aligned atoms.
- ☐ moving electric charge.
- ☐ none of the above
- ☐ tiny pieces of iron.



Question 25 3.5 pts

A straight 1.0-m long wire is carrying a current. The wire is placed perpendicular to a 0.20-T magnetic field B field. If the magnetic force exerted on the wire has a magnitude of 0.60 N, what is the current in the wire?

- ☐ 4.0 A

☐

3.0 A

☐

2.0 A

☐

5.0 A

☐

1.0 A



Question 26 3.5 pts

Magnet A has twice the magnetic field strength of Magnet B and at a certain distance pulls on magnet B with a force of 100 N. The amount of force that magnet A exerts on magnet B is

(hint: think Newton)

☐

exactly 100 N.

☐

need more information

☐

at or about 50 N.



Question 27 3.5 pts

The shape of a magnetic field surrounding a current-carrying conductor is

☐

neither of these

☐

all of these

☐

circular.

☐

radial.

☐

consistent with the inverse-square law.



Question 28 4 pts

A straight 1.0-m long wire is carrying a current. The wire is placed perpendicular to a magnetic field of strength 0.20 T. If the wire experiences a force of 0.60 N, what is the current in the wire?

☐

2A

☐

5A

☐

4A

☐

3A

☐

Question 29 4 pts

A 2.0-m straight wire carrying a current of 0.60 A is oriented parallel to a uniform magnetic field of 0.50 T.

What is the magnitude of the magnetic force on it?

☐

0.60N

☐

0

☐

0.15N

☐

0,30

Not saved

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